



Assessing the value of deep reinforcement learning for irrigation scheduling

T.D. Kelly^a, T. Foster^{a,*}, David M. Schultz^{b,c}

^a School of Engineering, University of Manchester, Manchester, UK

^b Department of Earth and Environmental Sciences, University of Manchester, Manchester, UK

^c Centre for Crisis Studies and Mitigation, University of Manchester, Manchester, UK

ARTICLE INFO

Keywords:

Water
Irrigation
Optimisation
Deep learning
Economic
Scheduling

ABSTRACT

Due to increasing global water scarcity pressure, researchers, policy makers and industry are looking for innovative solutions to increasing agricultural water productivity. Motivated by recent success within complex decision-making environments, Deep Reinforcement Learning (DRL) is being proposed as a method for optimizing irrigation strategies. Early research has hinted towards increased profits with DRL compared to heuristic approaches such as soil-moisture thresholds or fixed schedules. However, an assessment of the value of DRL for irrigation scheduling that incorporates local climate variability and water-use restrictions has yet to be performed. To address this gap in the literature, we created *aquacrop-gym*, an open-source Python framework for researchers to train and evaluate customized irrigation strategies within the crop-water model AquaCrop-OSPy. In this analysis, *aquacrop-gym* was used to quantify the value of DRL in comparison to conventional irrigation scheduling techniques (e.g., optimized soil-moisture heuristic) for maize production in an intensively irrigated region of the central United States. The DRL and heuristic approaches were both trained on 70 years of weather data produced from the weather generator LARS-WG, and evaluated on 30 unseen validation years of generated weather data. Findings from this analysis show that in the presence of high rainfall variability, DRL does not outperform conventional optimized heuristics. However, in the scenario where rainfall is set to zero, DRL approaches achieve higher profits on the unseen validation years. Similarly, DRL approaches also outperform optimized heuristics when severe water-use restrictions are introduced. Our analysis demonstrates that DRL approaches are a promising method of irrigation scheduling, notably in regions where farmers are faced with significant physical or regulatory water scarcity.

Introduction

Agriculture is the main consumptive user of freshwater in many water-scarce regions, and thus a sector that is both a cause and a victim of water scarcity. Improving agricultural water productivity is therefore an essential component of efforts to address water scarcity worldwide, as well as meeting food insecurity challenges [11,13].

Efforts to improve agricultural water productivity are fundamentally focused on tailoring irrigation to the crop-water needs as well as minimizing non-beneficial losses of water. Effective scheduling of irrigation during the growing season, alongside other interventions such as improved irrigation technologies or land management practices, are critical to achieving improvements in agricultural water productivity. Commonly, farmers schedule irrigation based on rules or heuristics, such as soil-moisture levels or crop conditions [35], which may be constant throughout the season or vary depending on crop growth stage.

However, while irrigation strategies based on simple rules and heuristics are an intuitive and agronomically logical way to schedule irrigation, they may lead to sub-optimal irrigation decisions if the underlying assumptions on which the heuristic is designed change or vary unexpectedly. For example, an unexpected rainstorm or longer than usual period of drought may render farmers' fixed irrigation heuristic sub-optimal in a given year. Similarly, an unanticipated change in water availability (e.g., due to equipment failure, emergency drought measures) may mean that farmers irrigation heuristics are now sub-optimal given altered water supply risks.

One solution to this challenge is to adapt irrigation scheduling heuristics by re-optimizing decision rules and triggers at various points within the season as the farmer begins to observe more of the seasonal weather conditions. Research has demonstrated that combining crop simulation models with field monitoring, weather forecast, and other data can be used to support adaptation of irrigation scheduling and

* Corresponding author.

E-mail address: timothy.foster@manchester.ac.uk (T. Foster).

<https://doi.org/10.1016/j.atech.2024.100403>

Received 10 November 2023; Received in revised form 15 January 2024; Accepted 15 January 2024

Available online 20 January 2024

2772-3755/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

improve yields, water productivity, and profitability [7,15,18,37]. However, this approach adds additional computational overhead during the season, especially if re-optimizing frequently throughout the season, thereby increasing costs for producers or irrigation scheduling service providers. In the case of service providers or farmers managing large land areas, re-optimizing strategies for tens of thousands of different cropping systems in near real-time may not be economically or technically feasible. Furthermore, as well as these issues with in-season adaptation, incorporating the increasing amounts of data available to farmers into heuristics will require large numbers of complex, interconnected rules that will be difficult to design and optimize. For example, it is not obvious how best to build an irrigation heuristic around short-term weather forecasts (e.g., irrigate if no rain is forecasted for the next five days), and how to combine this data with soil-moisture or canopy cover measurements to make the best decision possible.

Addressing these issues, artificial intelligence based approaches such as Deep Reinforcement Learning (DRL) [33] have been suggested as way to automatically and flexibly adjust and adapt decisions during the season in response to any relevant information. DRL approaches have successfully surpassed the best human benchmarks in other complex decision making environments such as video and board games [5,23,31,36]. Recent research has explored the viability of using DRL for irrigation scheduling [2,8,38]. In comparison with non-optimized schedules and heuristics, DRL methods were found to deliver improvements in water productivity and profits, highlighting their potential value as a tool for real-world irrigation management. However, in all of these works the DRL agent was evaluated on a limited number of unseen test seasons (less than 3 for all the studies referenced above), which is insufficient to assess performance across diverse and uncertain weather scenarios faced by farmers in real-world production environments. Additionally, trained DRL agents were compared only to handpicked schedules (i.e., a set of dates and depths chosen by researchers), fixed time intervals (i.e., irrigate 25 mm every N days), or single value heuristics (i.e., irrigate if the soil-water content drops a specified value), where this value was chosen by researchers rather than optimized for the specific production environment. These choices of benchmark irrigation strategy limit our understanding of the added value of using DRL compared to the optimized heuristics commonly used in irrigation scheduling research [16,19,20].

In this article, we conduct a more systematic assessment of the value of DRL approaches to irrigation scheduling in comparison to conventional optimized heuristics, across a range of climate conditions. We assess how design choices, such as when and how often the DRL agent makes irrigation decisions, as well as assumptions about water use restrictions, impacts estimates of productivity and profitability gains from DRL. Through these analyses, we provide new insights into how best to evaluate the benefits of DRL for irrigation scheduling, as well as discussions about the conditions under which DRL may be a powerful tool in improving irrigation water productivity.

Method

In this section we describe the framework used to train and evaluate DRL agents for irrigation scheduling. First, we give a brief overview of DRL (Section 2.1), then describe how we apply DRL to irrigation scheduling within a crop-simulation model (Section 2.2). The chosen case study, as well as experimental details are then described (Section 2.3), followed by a discussion on overfitting and how to report DRL performance (Section 2.4). Finally, we describe the optimized heuristic that we will use to assess the added value of DRL (Section 2.5).

Deep reinforcement learning

A typical reinforcement learning problem is framed as the interaction between an agent and an environment (Fig. 1). At each timestep, the agent receives some observation of the environment O , and makes action A in response. This action has some impact on the environment which then returns a reward R along with the next observation O . This loop continues until a terminal state of the environment is reached, after which point the *episode* has finished. The goal of the reinforcement learning agent is to learn a mapping of observation to action that maximizes the sum of future rewards [33].

In Deep Reinforcement Learning (DRL), the mapping of observation to action is an artificial neural network, with the weights of that neural network being continually updated in response to the rewards received by the agent. In this analysis we use Proximal Policy Optimization (PPO) [29], which is one of the most popular DRL algorithms and has been used in a wide variety of environments such as robotics [3], video games [23] and flow control [24]. PPO also has the advantage that it can be easily used for both continuous and discrete action spaces, allowing us to define many different types of irrigation strategies. Early experimentation found PPO provided the fastest and most stable training performance for this analysis compared to other algorithms tested. The code implementation of PPO that we use is from the DRL Python library *ray-rl* [17]. This library is an actively maintained, production-ready framework that is currently used within several industries (<https://docs.ray.io/en/latest/rllib/index.html>).

Applying DRL to irrigation scheduling

To train DRL agents to make irrigation decisions, a crop-simulation environment is needed to rapidly test different irrigation strategies. Moreover, this environment needs to follow the *Observation, Action, Reward, next Observation* (O–A–R–O) framework outlined in the previous section. For this analysis, we use AquaCrop-OSPy [16] to represent the cropping system, and serve as the environment with which the DRL agent can interact with, make irrigation decisions, and receive feedback on how these decisions impact seasonal profits. AquaCrop-OSPy is an open source Python implementation of the crop-water productivity

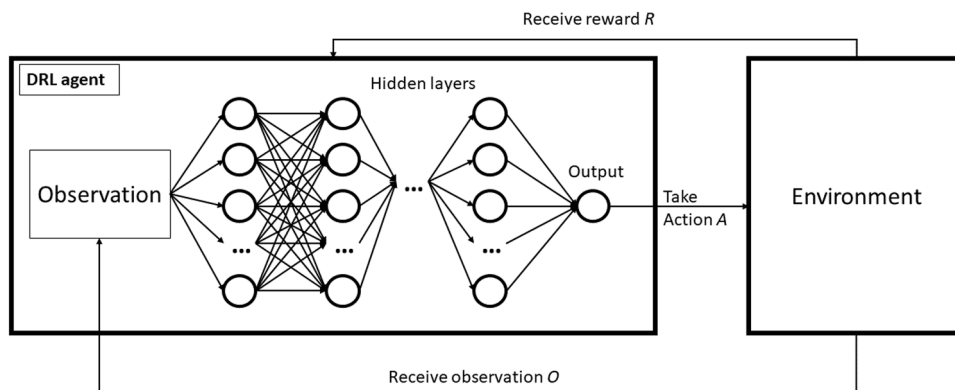


Fig. 1. Typical framing for a reinforcement learning problem, where a DRL agent aims to maximize the sum of future rewards.

model AquaCrop [32]. Developed by the UN-FAO, AquaCrop is one of the most widely used crop simulation models available, featuring in over 334 peer reviewed publications in the ten years since its launch [26]. AquaCrop aims to accurately represent crop response to water stress, whilst also reducing complexity and input data requirements, enabling it to be used in data scarce environments. AquaCrop-OSPy has been designed to allow flexible and complex irrigation decisions to be made outside of the model, allowing straightforward integration with DRL Python libraries or optimization algorithms. To facilitate this interaction between DRL agent and AquaCrop-OSPy, we created the *aquacrop-gym* open-source Python library (<https://github.com/aquacropos/aquacrop-gym>).

To train a DRL agent within *aquacrop-gym*, a DRL agent is first initiated with random neural network weights. To gather information with which to learn from, the agent then performs a number of *training rollouts* (Fig. 2). For each *training rollout* a weather year is randomly sampled from the training set and an AquaCrop-OSPy simulation is initialized. Every N number of days within the simulation, AquaCrop-OSPy will pass to the agent a set of observations (e.g., current soil-moisture content). These observations are then processed by the agent and an action is recommended (e.g., irrigation depth). The simulation then runs for another N days after which the agent will output another action based on the latest observation. At the end of the season, the final reward (e.g., seasonal profits) is calculated and sent back to the agent.

After the *training rollout* is complete, all of the saved (O–A–R–O) transitions are used to update the weights of the agent's internal neural network. Once the agent's strategy has been updated, a *validation rollout* (Fig. 2) is performed. The *validation rollout* evaluates the agent against all of the unseen validation years and calculates the mean profit achieved by the agent. Since none of the (O–A–R–O) transitions are saved during this *validation rollout*, no learning occurs and only the mean reward is saved to track the learning progress of the agent. This loop of: collect training data, update strategy, validate strategy will continue for a specified number of iterations or until manually stopped.

Case study, DRL design choices and experimental details

For the case study for this analysis, we use AquaCrop-OSPy to simulate irrigated maize production in southwest Nebraska. Maize is the most commonly grown and irrigated crop within Nebraska, which itself contains more irrigated acres than any other state in the United States [35]. Irrigation water in Nebraska is primarily sourced from the High Plains aquifer, which has experienced severe declines in aquifer storage as a result of over abstraction [22,28]. In many parts of Nebraska, farmers' water use is restricted by local management agencies to minimize groundwater over-pumping and associated externalities such as streamflow depletion [39].

To train and validate our model, we utilize historic weather data for southwest Nebraska from a monitoring station in Champion, Nebraska [14]. This weather station provides 37 years (1982–2018) of observational data (daily minimum and maximum temperature, precipitation, and solar radiation), which we subsequently input to the weather generator LARS-WG [30] to generate a longer 100-year synthetic time series for our analysis. This weather generator determines the statistical characteristics of a given daily weather time series and uses these characteristics to generate new time series of daily weather data with the same statistical properties in terms of weather variability within and across years. LARS-WG has been shown in multiple recent studies to successfully represent the weather characteristics (e.g., monthly mean and variance of temperature or precipitation) of the input time-series [1, 4]. Using this method, we greatly increase the number of weather years for training (70 years) and validating (30 years) whilst still adequately representing the observed historical weather conditions (Figure S1). Other inputs, including crop and soil parameters, for AquaCrop simulations are drawn from the literature, specifically from previous studies that use the model to simulate maize production in the central United States [12,20,27]. For calculation of seasonal profits at the end of each simulation, we assume a crop price of \$180 per tonne [34], irrigation costs of \$1 per ha-mm [9], and fixed non-irrigation production costs of \$1728 ha⁻¹ [9]. Main DRL hyperparameters are specified in Table S1, and full details of source code, model and DRL parameters are all available at (<https://github.com/aquacropos/aquacrop-gym>). We focus

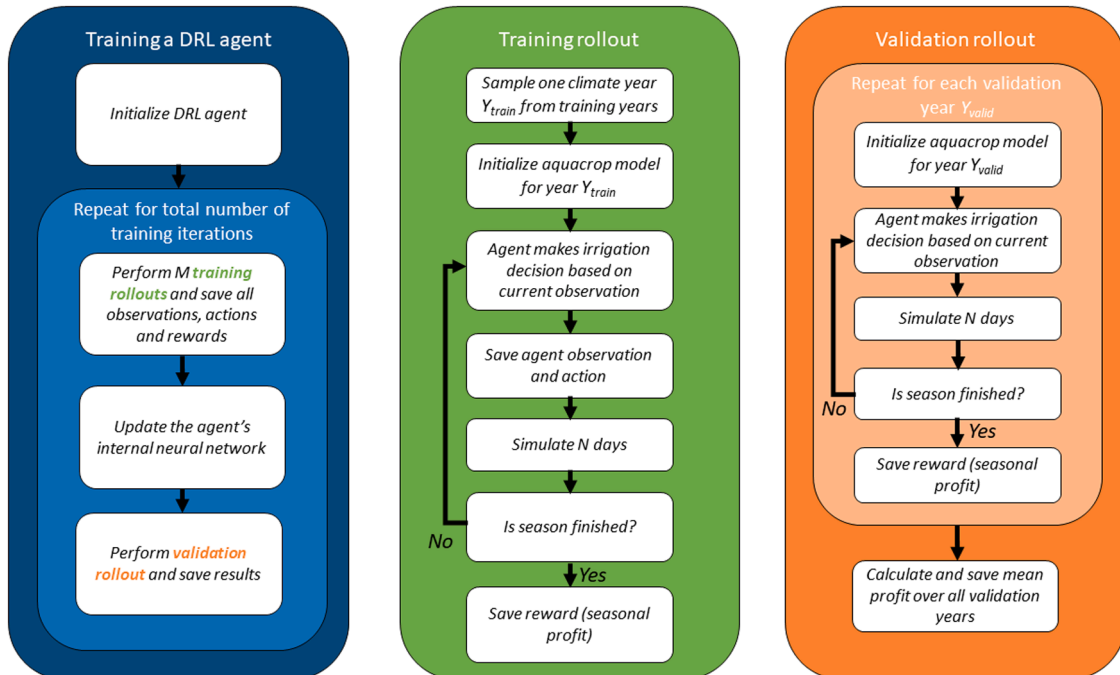


Fig. 2. Flow diagram detailing how training data is collected by the DRL agent, then used to update the agent's decision making. After updating the new strategy is evaluated on the validation set.

on seasonal profit as our metric of performance as profit represents the primary motivation of farmers operating in high-productivity technologically-advanced farming systems such as the central United States. Moreover, maximisation of profit also provides greater opportunity (e.g. compared to maximisation of yield) to assess benefits of DRL scheduling, due to the complexity of managing trade-offs between costs of water use and economic returns from crop yield.

Determining what observations the agent should receive, which actions it is allowed to take, and what reward it will maximize are critical design choices that will depend on the cropping system being simulated. Examining all possible combinations of observation and action sets is out of the scope of this analysis. However, our aim is to test several of the most common and likely combinations that are applicable for our study area in the central United States. The range of action sets chosen for our analysis are displayed in Table 1. For *depth discreet*, the agent chooses an irrigation depth to apply on the following day from a set of choices (0, 2.5, 5, ..., 22.5, 25 mm ha⁻¹). For *depth continuous*, the agent can choose any depth to apply the following day within the specified bounds [0, 25 mm ha⁻¹]. For *binary*, the agent decides whether or not to apply 25 mm of irrigation the following day. For *soil-moisture threshold*, the agent decides the soil-moisture threshold to use until the next decision step. If the soil-moisture content ever drops below this threshold, irrigation is triggered to fill the soil back to field capacity (subject to the maximum irrigation application of 25 mm ha⁻¹). Finally, for all of these action sets, we compare four different decision intervals (i.e., the number of days between actions) 1, 3, 5 and 7 days.

In each case, observations provided to the agent before each irrigation decision are as follows: mean daily reference evapotranspiration and precipitation for the season so far; and for the past 7 days only, current root-zone soil-moisture content, canopy cover, biomass growth, cumulative irrigation, water supply remaining, crop growth stage, and the total number of days since the season began. These variables are either obtained through the input weather time series or outputted by the AquaCrop-OSPy model. This choice of variables represents a range of weather, soil, and crop condition information that could plausibly be monitored or collected by farmers to support a DRL based scheduling approach, and could be extended or reduced depending on the monitoring capabilities of the individual production system being modelled.

As discussed in the previous section, DRL agents will be trained on 70 years of training data and evaluated on 30 years of validation data. The mean and standard deviation in seasonal rainfall (May–September) for the training and validation sets were closely comparable, equal to 309.6 ± 93.7 and 306 ± 93.2 mm ha⁻¹ respectively. Although both the training and validation sets were drawn from the same statistical distribution, the individual time series will be different, and thus provides a way for us to assess how well the agent might perform if deployed in the real world under uncertain weather conditions. To assess the impact of this local variability on DRL performance, an additional experiment was therefore conducted where the rainfall was removed completely from the training and validation sets. We hypothesise that by removing this important source of weather variability, and hence reduce the difference between the training and validation time series, the performance of DRL should improve compared to when rainfall is present.

Previous assessments of DRL for irrigation scheduling have assumed unrestricted water supplies for irrigation, whereby the DRL agent can irrigate as much as required. However, this assumption is not realistic to our case study region of Nebraska, or even other regions globally where

water use may be restricted due to regulatory policies such as yearly or multi-yearly quotas and allocations or physical supply and capacity constraints [6,39]. Given that the marginal gains from applying additional water are greatest in times of water scarcity, we hypothesise that more complex strategies such as DRL may be more valuable when water resources are limited. To assess the impact of water use restrictions on the value of DRL, the framework presented so far is repeated and water use limits of 75, 100, 125 mm ha⁻¹ are imposed on all simulation years. This restriction will prevent any further irrigation once the cap has been reached in any given growing season.

To summarise, four DRL agents will be trained initially to compare the action sets described in Table 1. The best performing action set will be used to train an additional four DRL agents to compare the four decision intervals 1, 3, 5 and 7 days. With this chosen DRL configuration (i.e., action set and decision interval), two more experiments will be conducted – one with no change to local rainfall and one with rainfall set to zero. Finally, with the same DRL configuration, three experiments will be conducted for each seasonal water use limit 75, 100, 125 mm ha⁻¹. In total this will result in 13 experiments.

Learning curves, evaluation criteria and overfitting

One of the downsides of using an artificial neural network based irrigation strategy is that neural networks, if trained for long enough, will often overfit to training data. When overfitting occurs, the training set performance continues to increase but the performance on the unseen validation set begins to decrease. This divergence occurs as the neural network begins to find patterns that apply only to the training set, rather than general patterns that apply also to the validation set. As a result of this overfitting, it is often desirable to stop training before reaching an optimum score on the training set – by which point the model has very likely overfit. To track learning process, throughout training the DRL agent will be continually evaluated on both the training and validation years, allowing us to monitor when and if overfitting occurs.

Fig. 3 displays these evaluation results for a DRL agent that makes *discreet depth* decisions every 7 days. Fig. 3a is generally referred to as the learning curve and tracks the agent's performance as a function of training steps (i.e., number of irrigation decisions made and learned from). Performance on the training set continues to increase throughout. However, the validation set performance declines after approximately 500,000 training steps. The divergence between training and validation performance highlights an important decision that needs to be made regarding how to report the final DRL performance, with which to compare against other irrigation strategies. Once choice is to report the highest validation profits achieved by the agent, which would give an indication of the potential of DRL if training is stopped at the right moment. However, this choice would overestimate the performance of DRL as it assumes knowledge of the validation years, which would not be available to a farmer as the validation set represents future unknown weather. Another choice is to use the final validation set profit after training has stopped, which would not assume any knowledge of the validation years. However, as shown in Fig. 3a, there can be large variation in validation profits between learning updates, and so choosing just one evaluation result may be misleading. Therefore, we instead calculate the rolling mean across the previous 10 validation set evaluations (Fig. 3b). When comparing the performance of DRL on the validation set, we will report the highest 10-evaluation average.

Comparison with an optimal benchmark irrigation heuristic

Previous assessments of DRL for irrigation scheduling have only compared DRL agents with handpicked schedules or one-value non-optimized heuristics [2,8,38]. A common irrigation scheduling heuristic used by farmers in this region is to schedule irrigation based on soil-moisture target thresholds. In this approach, irrigation is triggered

Table 1

The action sets this case study compares.

Action set	Bounds	Example action
<i>discreet depth</i>	[0,25] mm ha ⁻¹	0, 2.5, ... 22.5, 25 mm ha ⁻¹
<i>continuous depth</i>	[0,25] mm ha ⁻¹	8.43 mm ha ⁻¹
<i>Binary</i>	[0,25] mm ha ⁻¹	0, 25 mm ha ⁻¹
<i>soil moisture threshold</i>	[0,100]% TAW	63.2 %TAW

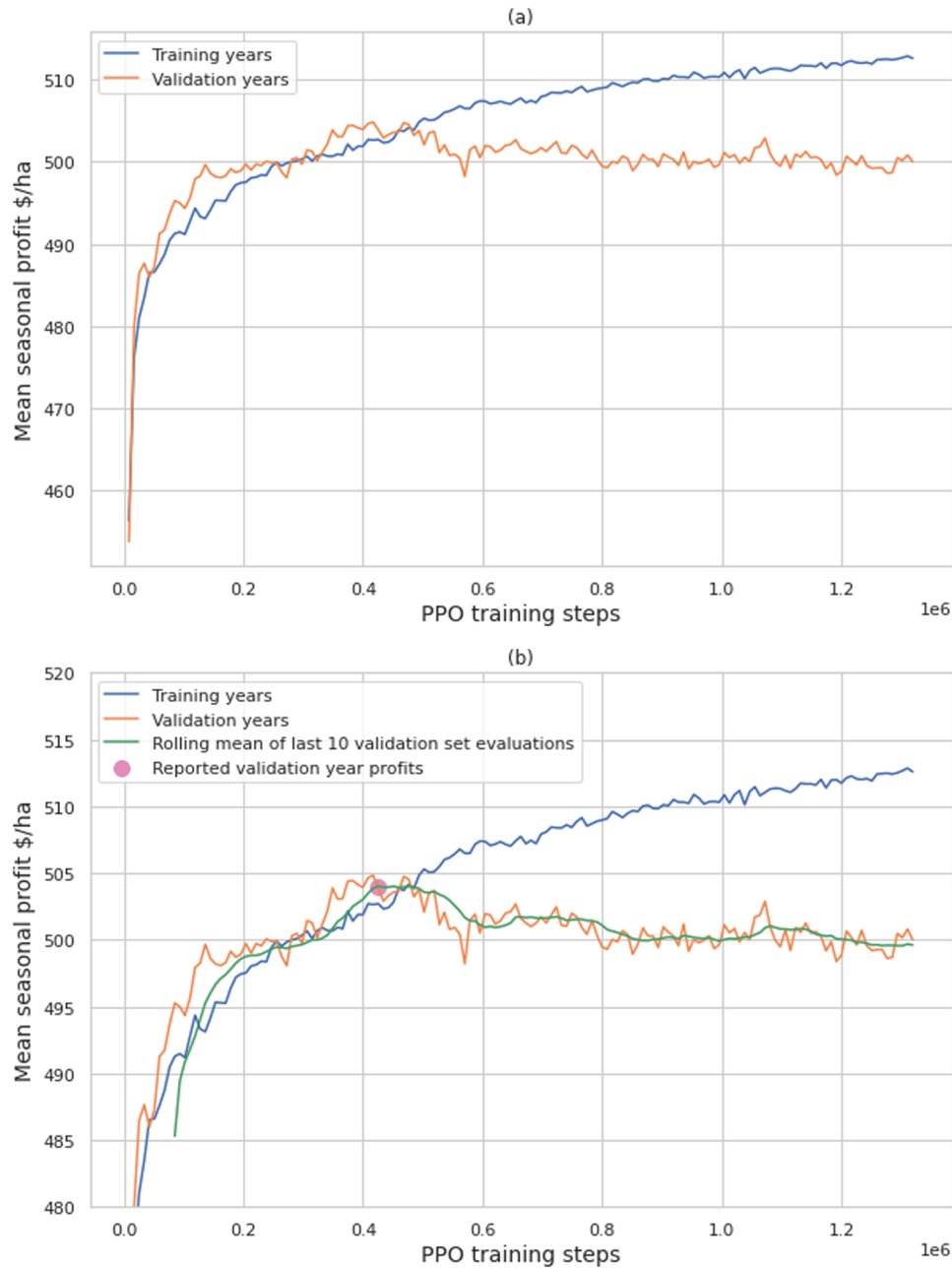


Fig. 3. (a) Average training profits and testing profits as a function of training steps. (b) Average training profits and testing profits as a function of training steps, as well as a rolling average of validation profits over 10 evaluations. The highest point on this rolling average calculation will be reported as the validation profit.

on any day when the soil-water content drops below a threshold level, which itself can be varied to account for the changing sensitivity of a crop to water deficits during the season. As a benchmark to compare our DRL agent against, we consider a farmer who uses a set of four soil-moisture target thresholds corresponding to the four main growth stages within AquaCrop. We use *aquacrop-gym* to identify to define a set of four soil-moisture thresholds that maximize average profit over the 70 training years. This set of thresholds will then be evaluated against the unseen 30 validation years and thus provide the benchmark against which to assess the added value of DRL. This benchmark achieves \$513 ha^{-1} on the training years and \$521 ha^{-1} on the validation years. The benchmark also achieves \$503 ha^{-1} when evaluated on the 37 years of historical data that was originally imputed into LARS-WG. By comparing these profits to those that could be achieved by directly optimizing the heuristic on those 37 historical years (\$506 ha^{-1}), we can be confident that our approach of using generated data to train and optimize

strategies – as well as the choice of weather generator – produces robust strategies for the case study region.

Optimising the heuristic benchmark as well as running DRL training experiments were all conducted on a single AMD Ryzen 7 3700X, 3.60 GHz, 8 core computer, with a GeForce RTX 3070 GPU. Training 1 million DRL training steps on this machine took approximately 4 h, whereas optimizing a set of four soil-moisture thresholds took approximately 20 min.

Results

This section begins by assessing how the performance of DRL varies depending on choices of action sets and decision frequencies (Section 3.1). Next, the effect of climate variability on DRL performance will be assessed (Section 3.2), followed by the introduction of restrictive caps on irrigation water use (Sections 3.3).

Comparing DRL configurations

Table 2 shows the mean seasonal profits achieved by different irrigation scheduling methods after 1 million training steps. DRL approaches, independent of these choices for how irrigation could be triggered, achieved higher profits on the training set than the fixed optimized irrigation scheduling heuristic. However, the fixed heuristic achieved marginally higher profits than the DRL approaches over the validation set. This result illustrates that deep learning algorithms are able to learn complex irrigation strategies (as evidenced by the superior training set performance), but may not be able to easily transfer those superior strategies into improved performance in out-of-sample production scenarios (i.e., the validation set). The differences in profit between all strategies are extremely marginal and so these results alone cannot be used to definitively rule out any approach as a viable method of scheduling irrigation. The *discreet depth* action set achieved the highest validation profit of all DRL scheduling strategies, and so will be used as the basis for comparison with fixed soil-moisture threshold (SMT) heuristics for the rest of the analysis presented.

The second DRL design choice we examined was the decision interval (i.e., how often the DRL agent makes irrigation decisions). For the best performing action set (*discreet depth*), we train the DRL agent with decision intervals (1, 3, 5, 7 days) for 1 million training steps and compare the validation set performance. This comparison showed that a decision interval of 3 days resulted in the highest profits over both the training and validation set (Table 3). The variation in profits between different decision intervals was greater than the profit variations found between different action sets (Table 2). This result could indicate that how often the DRL agent makes decisions is more important than how the amount of irrigation to apply is determined when an irrigation event is triggered, at least for our specific crop model environment and case study.

Climate variability

Despite the range and distribution of weather scenarios in model training and validation being comparable, the results presented in Section 3.1 show that complex DRL scheduling algorithms have thus far only been able outperform simpler optimized soil-moisture thresholds on the training set. One reason for this underperformance is that specific weather scenarios encountered in the validation period are inherently different (e.g., in terms of specific sequencing of rainfall events), despite still having same overall statistical distribution (e.g., in terms of total seasonal rainfall). By setting the rainfall to zero, and hence reducing the difference between the training and validation time series, we expect the performance of DRL to improve on the validation set. To test this hypothesis, we removed the rainfall from both the training and validation weather dataset used within AquaCrop-OSPy, and retrained the DRL agent as well as re-optimized the fixed heuristic benchmark.

Table 4 shows the difference in profits between the DRL agent and the optimized soil-moisture thresholds when rainfall is included or removed. When rainfall was removed, optimized fixed soil-moisture thresholds achieved a profit of \$300.6 ha⁻¹ on the validation set, with reduced profit relative to that found in Section 3.1 due to larger irrigation amounts needed to meet crop water requirements across all years.

Table 2

Reported seasonal profits (highest mean profits over 10 evaluations) for different action sets.

Scheduling method	Training set profits \$ ha ⁻¹	Validation set profits \$ ha ⁻¹
DRL – <i>discreet depth</i>	518.9	520.1
DRL – <i>continuous depth</i>	520.2	512.6
DRL – <i>binary</i>	515.3	517.8
DRL – <i>soil-moisture threshold</i>	517.5	515.7
Fixed Optimized SMT	513.1	521.4

Table 3

Reported seasonal profits (highest mean profits over 10 evaluations) for different decision intervals using the *discreet depth* action set defined in Section 2.3.

Decision Interval	Training set profits \$ ha ⁻¹	Validation set profits \$ ha ⁻¹
1 day	492.2	497.6
3 days	518.9	520.1
5 days	515.1	511.0
7 days	510.4	504.0

Table 4

Increase in seasonal profits compared to optimized soil-moisture thresholds with and without rainfall.

Weather data	Increase in training set profits \$ ha ⁻¹	Increase in validation set profits \$ ha ⁻¹	Percentage increase in validation set profits
With rainfall	5.87	−1.27	−0.2 %
Without rainfall	4.07	6.38	+2.1 %

In contrast, the DRL agent taking *discreet depth* actions every 3 days achieved \$307 ha⁻¹. This result indicates that removing one of the main drivers of variability and uncertainty affecting farmers irrigation decisions altered the added value of DRL scheduling relative to fixed SMT heuristics from a reduction of \$1.27 ha⁻¹ to an increase of \$6.38 ha⁻¹, the latter a 2.1 % increase in profits compared to the optimized heuristic.

Water use restrictions

The DRL and heuristic strategies evaluated so far have operated under unrestricted irrigation, whereby the DRL agent can irrigate as much as required. However, this assumption is not realistic to our case study region of Nebraska, which imposes yearly or multi-year restrictions on irrigation water abstractions. Given that the marginal gains from applying additional water are greatest in times of water scarcity, we hypothesise that more complex strategies such as DRL may be more valuable when water resources are limited. To assess the impact of water use restrictions on the value of DRL, the framework presented so far was repeated and water use limits of 75, 100, 125 mm ha⁻¹ were imposed on all simulation years. This restriction will prevent any further irrigation once the cap has been reached in any given growing season. Table 5 shows the difference in profits between the DRL agent and the optimized soil-moisture thresholds for these different water use restrictions. Despite only small profit increase, DRL does provided added value in these water restricted scenarios compared to optimized heuristics. By introducing the most restrictive cap evaluated (75 mm ha⁻¹), the value of DRL increased from −\$1.3 ha⁻¹ to +\$12.2 ha⁻¹, which equates to a 5.6 % increase in profits compared to the optimized heuristic.

Discussion

Despite the success of optimized heuristics within irrigation scheduling, incorporating the increasing amounts of data available to farmers

Table 5

The effect of water use restrictions on validation set profits and added value of DRL compared to optimized heuristics.

Irrigation cap mm ha ⁻¹	DRL validation set profits \$ ha ⁻¹	Increase in validation set profits \$ ha ⁻¹	Percentage Increase in validation set profits
75	228	12.2	5.6 %
100	286.5	7.1	2.5 %
125	332	4.3	1.3 %
No cap	520.1	−1.3	−0.2 %

into heuristics will require large numbers of complex, interconnected rules that will be difficult to design and optimize. Addressing these issues, artificial intelligence based approaches such as Deep Reinforcement Learning (DRL) [33] has been suggested as way to automatically and flexibly adjust and adapt decisions during the season in response to any relevant information. Recent publications have found Deep Reinforcement Learning (DRL) can be used to learn irrigation strategies within crop-simulation frameworks [2,8,38].

In this article we assessed the added value of using DRL for irrigation scheduling in comparison to optimized soil-moisture thresholds. The first part of the analysis found that allowing the DRL agent to choose between a range of discrete irrigation depths (i.e., 0, 2.5, ... 22.5, 25 mm) with frequent irrigation (every 3 days) lead to the highest profits when assessed in validation years. More (e.g. 1 every day) or less (e.g. every 5 or 7 days) frequent irrigation decision intervals resulted in marginally lower crop profits. We attribute this finding to the fact that daily irrigation decisions create the most complex irrigation schedules, increasing potential for maladaptive irrigation decisions (e.g. irrigation shortly before a rainfall event) to be made by the DRL agent as the irrigation schedule solution space grows, whereas longer irrigation decision intervals increase risks of crop water stress occurring between days when the DRL agent makes choices about irrigation scheduling.

Despite performing the best compared to other DRL design choices, this was still not enough to surpass the profits achieved by a conventional optimized irrigation heuristic in the same validation years. When rainfall was removed from the weather data, thereby decreasing the difference in weather time series between training and validation sets, the DRL agent was now able to increase validation year profits by 2.1 % compared to optimized heuristics. Finally, when restrictive irrigation caps were introduced (e.g., 75 mm ha⁻¹), the DRL agent was able to increase validation year profits by 5.6 % compared to the optimized heuristics.

Our analysis found that under normal climate variability for the case study region and unrestricted water use, optimized soil-moisture thresholds achieved a higher average profit over the validation years than the DRL agent. This result broadly agrees with [2]), who found that even a non-optimized fixed heuristic achieved higher profits than their DRL agent during one of their two test years. This marginal difference in validation year profits (from our analysis) found between DRL and optimized heuristic (– \$1.3 ha⁻¹) cannot be taken as an exact quantitative comparison on its own, as another set of validation years could cause this value to vary. Differences between how DRL performance is reported – i.e., when training is stopped, how validation set profits are calculated, and how many training repetitions are performed – can also lead to this difference in profits to vary slightly. However, the qualitative result that DRL agents struggle to surpass optimized thresholds is likely to remain, an insight that has been largely neglected in previous DRL assessments [2,8,38]. This finding that fixed soil-moisture heuristics perform well compared to DRL – given the simplicity and intuitive nature of these heuristics – should indicate that fixed optimized heuristics are good first step towards increasing productivity that producers would be more likely to adopt and implement.

Our finding that the value of DRL increases with the introduction of restrictive water use caps provides important insights about where DRL is best implemented in real world cropping systems. As water resources become scarce and demands over them increase [10,25], water managers and policy makers may introduce restrictions on total water use – as some regions already do [6,39]. Our analysis indicates that DRL may help farmers to more effectively schedule limited irrigation water supplies as water restrictions become more stringent and variable, with producers potentially being able to use these intelligent methods to mitigate impacts of increasing water insecurity. Our other finding that increased value in DRL for no-rain scenarios implies that DRL may also be better placed in arid areas where there is very little rainfall, and thus more limited differences in crop irrigation demands and their intra-seasonal distribution from year to year.

Although we have been able to evaluate different action sets, decision frequencies, water use restrictions and climate variability, computational constraints mean we have only been able to evaluate some combinations of these factors in this paper. For example, we assess the effects of irrigation decision intervals for discrete depth actions, but further work should also explore the interactions between irrigation intervals and other scheduling actions or constraints. Additionally, we have neglected comparing other important design choices such as different DRL algorithms, hyperparameters, neural network architectures and input observations. For example, using larger neural networks or expanding the number of input observations will lead to more complex irrigation strategies, and hence potentially increased profits, though at the expense of longer training times and higher risk of over-fitting. Investigating the effect of these and other DRL design choices in the value of DRL will be an important area of future work.

Applying DRL to irrigation scheduling, as we have done in this analysis, presents several issues for DRL that are not seen to the same extent in the typical environments DRL has been previously applied in. Assuming deployment within a real-world field, the agent will have to make decisions for several months or even years before receiving a reward signal (e.g., profits calculated after harvesting). This is because the outcome (e.g. crop production or profit) from an individual irrigation event is not immediately observed, with a DRL agent only observing an outcome and associated reward at harvest based on the combined effects of multiple irrigation decisions or choices during the growing season. This issue is partly circumvented by training the agent within crop-simulation environments, which can simulate whole seasons within seconds. However, the costly and time-consuming work of field trials will still have to be conducted eventually to test if model-optimized strategies will work as intended. This challenge of transferring simulated learning to real-world systems is receiving attention within robotics research [40]. However, the months needed to evaluate a single irrigation strategy is unique to irrigation scheduling. Another inherent difficulty of applying DRL to irrigation scheduling is that there are an arbitrarily large number of environment configurations (e.g., crop, soil, climate, management constraints) that dramatically impact optimal decision-making. This issue either requires separate DRL agents to be trained on each configuration – which may be too computationally expensive or time consuming, as with optimized heuristics – or instead training one agent to make decisions for all configurations – which may be too difficult to learn good strategies. Moreover, for farmers to adopt a DRL irrigation strategy, the added value will have to be demonstrated for their particular configuration, the value of DRL may have to be demonstrated in the field for many configurations in order for producers to adopt this new scheduling approach. Therefore, this issue of configuration agnostic DRL irrigation strategies will be an essential area of future work.

Similarly, the irrigation scheduling problem specified in this work, and much of the previous literature [2,7,8,15,21,38], centres around scheduling irrigation for a single homogenous field over a number of seasons. This specification, although simple to implement, does favour an optimized heuristic approach (i.e., optimization algorithms can comfortably find a set of values that maximize average profits in one field over a set of N seasons). However, heuristics do not scale well with increasing complexity, for example, if the seasonal water limit changes during the season, the fixed heuristics would have to remain sub-optimal or be re-optimized each time the situation changes. Another example is in the case of multiple fields or crops, where decisions may have to be made about which field or crop to prioritize with the limited water supply available. Finally, an example from our case study region is dealing with a 5-year allocation of irrigation water supply, meaning decisions must be made in the knowledge of how much water needs to be kept for future years. In these more complex – and more realistic – modelling scenarios, it is possible that DRL will demonstrate increased additional value.

Conclusions

Due to the increase in water scarcity issues globally, researchers are looking for innovative solutions to increase agricultural water productivity. One of the methods being investigated is Deep Reinforcement Learning (DRL) which has had recent success inside complex decision-making environments. In this article we assess the value of DRL for irrigation scheduling in comparison to optimized heuristics for a case study of irrigated maize production in southwest Nebraska. By designing a framework for DRL agents to interact with the AquaCrop-OSPy crop-simulation model, DRL agents were trained and evaluated over separate sets of climate data. Under normal climate variability and unrestricted irrigation, DRL failed to increase profits on the validation years compared to optimized soil-moisture thresholds. When rainfall was removed, and hence the biggest source of climate variability, DRL was able to surpass the performance soil-moisture thresholds. With the addition of restrictive irrigation caps, DRL was also able to achieve higher profits on the validation years than optimized thresholds. Our work firstly highlights that DRL is a promising avenue for improving irrigation scheduling, but DRL's additional value over optimized heuristics has yet to be demonstrated for all cases, especially given their increased complexity and data requirements. Conversely, optimized heuristics – given their relative simplicity – should be seen as an important first step to improving water productivity, before making large investments in AI based approaches.

Ethics statement

Not applicable: This manuscript does not include human or animal research.

CRediT authorship contribution statement

T.D. Kelly: Conceptualization, Data curation, Formal analysis, Methodology, Software. **T. Foster:** Conceptualization, Funding acquisition, Methodology, Software. **David M. Schultz:** Funding acquisition, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All code used in the analysis are available in a public github repository (<https://github.com/aquacropos/aquacrop-gym>). All input data are publicly available and sources cited in the manuscript.

Acknowledgments, Samples, and Data

The work contained in this article was funded by the National Environmental Research Council's Understanding the Earth, Atmosphere, and Ocean Doctoral Training Programme, Grant NE/L002469/1. To support research into these areas of future work, we created *aquacrop-gym*, an open-source python library that supports the training of DRL agents and comparisons between irrigation strategies within AquaCrop-OSPy. Training materials have been developed that guide the user to training their own DRL agents and duplicate the analysis presented in this work. The intention of this library is to be an actively developed framework that future researchers can test new irrigation strategies and compare them to the current best approaches. This framework will accelerate progress in optimized and intelligent irrigation scheduling research, as well as how rapidly this research can be

converted to production. All code used in the analysis are available in a public github repository (<https://github.com/aquacropos/aquacrop-gym>), allowing anyone to duplicate the analysis.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.atech.2024.100403](https://doi.org/10.1016/j.atech.2024.100403).

References

- [1] A. Agarwal, M.S. Babel, S. Maskey, Analysis of future precipitation in the Koshi river basin, Nepal, *J. Hydrol.* 513 (2014) 422–434, <https://doi.org/10.1016/j.jhydrol.2014.03.047>.
- [2] K. Alibabaei, P.D. Gaspar, E. Assunção, S. Alirezazadeh, T.M. Lima, Irrigation optimization with a deep reinforcement learning model: case study on a site in Portugal, *Agric. Water Manage.* 263 (2022) 107480, <https://doi.org/10.1016/j.agwat.2022.107480>.
- [3] Andrychowicz, O.A.M., Baker, B., Chociej, M., Józefowicz, R., McGrew, B., Pachocki, J., Petron, A., Plappert, M., Powell, G., Ray, A., Schneider, J., Sidor, S., Tobin, J., Welinder, P., Weng, L., & Zaremba, W. (2019). Learning dexterous in-hand manipulation: <https://doi.org/10.1177/0278364919887447>, 39(1), 3–20.
- [4] R. Awal, H.K. Bayabil, A. Fares, Analysis of potential future climate and climate extremes in the brazos headwaters basin, Texas, *Water* 8 (12) (2016) 603, <https://doi.org/10.3390/W8120603>, 20168603.
- [5] A.P. Badia, B. Piot, S. Kapturowski, P. Sprechmann, A. Vitvitskiy, D. Guo, C. Blundell, Agent57: outperforming the Atari Human Benchmark, in: Proceedings of the 37th International Conference on Machine Learning, ICML, 2020, pp. 484–494, <https://doi.org/10.48550/arxiv.2003.13350>, 2020, PartF168147-1.
- [6] J. Berbel, A. Expósito, C. Gutiérrez-Martín, L. Mateos, Effects of the irrigation modernization in Spain 2002–2015, *Water Resour. Manage.* 33 (5) (2019) 1835–1849, <https://doi.org/10.1007/s11269-019-02215-w>.
- [7] X. Cai, M.I. Hejazi, D. Wang, Value of probabilistic weather forecasts: assessment by real-time optimization of irrigation scheduling, *J. Water Resour. Plann. Manage.* 137 (5) (2011) 391–403, [https://doi.org/10.1061/\(asce\)wr.1943-5452.0000126](https://doi.org/10.1061/(asce)wr.1943-5452.0000126).
- [8] M. Chen, Y. Cui, X. Wang, H. Xie, F. Liu, T. Luo, S. Zheng, Y. Luo, A reinforcement learning approach to irrigation decision-making for rice using weather forecasts, *Agric. Water Manage.* 250 (2021) 106838, <https://doi.org/10.1016/j.agwat.2021.106838>.
- [9] CropWatch. (2019). Nebraska Crop Budgets. <https://cropwatch.unl.edu/budgets>.
- [10] P. D'Odorico, D.D. Chiarelli, L. Rosa, A. Bini, D. Zilberman, M.C. Rulli, The global value of water in agriculture, *Proc. Natl. Acad. Sci. USA* 117 (36) (2020) 21985–21993, https://doi.org/10.1073/PNAS.2005835117/SUPPL_FILE/PNAS.2005835117.SAPP.PDF.
- [11] E. Fereres, M.A. Soriano, Deficit irrigation for reducing agricultural water use, *J. Exp. Bot.* 58 (2) (2006) 147–159, <https://doi.org/10.1093/jxb/erl165>.
- [12] T. Foster, N. Brozović, A.P. Butler, Analysis of the impacts of well yield and groundwater depth on irrigated agriculture, *J. Hydrol.* 523 (2015) 86–96, <https://doi.org/10.1016/j.jhydrol.2015.01.032>.
- [13] T.A. Howell, Enhancing water use efficiency in irrigated agriculture, *Agron. J.* 93 (2) (2001) 281–289, <https://doi.org/10.2134/AGRONJ2001.932281X>.
- [14] HPRCC. (2016). High Plains Regional Climate Center. <https://hprcc.unl.edu/index.php>.
- [15] A. Jamal, R. Linker, M. Housh, Optimal irrigation with perfect weekly forecasts versus imperfect seasonal forecasts, *J. Water Resour. Plann. Manage.* 145 (5) (2019) 06019003, [https://doi.org/10.1061/\(ASCE\)WR.1943-5452.0001066](https://doi.org/10.1061/(ASCE)WR.1943-5452.0001066).
- [16] T.D. Kelly, T. Foster, AquaCrop-OSPy: bridging the gap between research and practice in crop-water modeling, *Agric. Water Manage.* 254 (2021) 106976, <https://doi.org/10.1016/j.agwat.2021.106976>.
- [17] Liang, E., Liaw, R., Moritz, P., Nishihara, R., Fox, R., Goldberg, K., Gonzalez, J.E., Jordan, M.I., & Stoica, I. (2017). RLlib: abstractions for distributed reinforcement learning. 35th International Conference on Machine Learning, ICML 2018, 7, 4768–4780. <https://doi.org/10.48550/arxiv.1712.09381>.
- [18] R. Linker, Stochastic model-based optimization of irrigation scheduling, *Agric. Water Manage.* 243 (2021) 106480, <https://doi.org/10.1016/j.agwat.2020.106480>.
- [19] R. Linker, I. Ioslovich, G. Sylaios, F. Plauborg, A. Battilani, Optimal model-based deficit irrigation scheduling using AquaCrop: a simulation study with cotton, potato and tomato, *Agric. Water Manage.* 163 (2016) 236–243, <https://doi.org/10.1016/j.agwat.2015.09.011>.
- [20] R. Linker, I. Kisekka, Model-Based deficit irrigation of maize in Kansas, *Trans. ASABE* 60 (6) (2017) 2011–2022, <https://doi.org/10.13031/trans.12341>.
- [21] R. Linker, G. Sylaios, I. Tsakmakis, T. Ramos, L. Simionesei, F. Plauborg, A. Battilani, Sub-optimal model-based deficit irrigation scheduling with realistic weather forecasts, *Irrigation Sci.* 36 (6) (2018) 349–362, <https://doi.org/10.1007/s00271-018-0592-x>.
- [22] McGuire, V.L. (2017). Water-Level and Recoverable Water in Storage Changes, High Plains Aquifer, Predevelopment to 2015 and 2013–15. U.S. Geological Survey Scientific Investigations Report 2017–5040, 14. <https://doi.org/10.3133/sir20175040>.
- [23] OpenAI, :, Berner, C., Brockman, G., Chan, B., Cheung, V., Debiak, P., Dennison, C., Farhi, D., Fischer, Q., Hashme, S., Hesse, C., Józefowicz, R., Gray, S., Olsson, C.,

- Pachocki, J., Petrov, M., Pinto, H.P. de O., Raiman, J., ... Zhang, S. (2019). Dota 2 With Large Scale Deep Reinforcement Learning.
- [24] J. Rabault, M. Kuchta, A. Jensen, U. Réglade, N. Cerardi, Artificial neural networks trained through deep reinforcement learning discover control strategies for active flow control, *J. Fluid Mech.* 865 (2019) 281–302, <https://doi.org/10.1017/JFM.2019.62>.
- [25] L. Rosa, D.D. Chiarelli, M.C. Rulli, J. Dell'Angelo, P. D'Odorico, Global agricultural economic water scarcity, *Sci. Adv.* (18) (2020) 6, https://doi.org/10.1126/SCIADV.AAZ6031/SUPPL_FILE/AAZ6031_TABLES_S2_TO_S5.XLSX.
- [26] Elias M. Salman, M. García-Vila, E. Fereres, D. Raes, P. Steduto, The AquaCrop model – Enhancing crop water productivity. The AquaCrop model – Enhancing crop Water Productivity, FAO, 2021, <https://doi.org/10.4060/cb7392en>.
- [27] R. Sandhu, S. Irmak, Performance of AquaCrop model in simulating maize growth, yield, and evapotranspiration under rainfed, limited and full irrigation, *Agric. Water Manage.* 223 (2019) 105687, <https://doi.org/10.1016/j.agwat.2019.105687>.
- [28] B.R. Scanlon, C.C. Faunt, L. Longuevergne, R.C. Reedy, W.M. Alley, V.L. McGuire, P.B. McMahon, Groundwater depletion and sustainability of irrigation in the US high plains and central valley, *Proc. Natl. Acad. Sci. USA* 109 (24) (2012) 9320–9325, <https://doi.org/10.1073/pnas.1200311109>.
- [29] Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Openai, O.K. (2017). Proximal Policy Optimization Algorithms. <https://doi.org/10.48550/arxiv.1707.06347>.
- [30] M.A. Semenov, E.M. Barrow, Use of a stochastic weather generator in the development of climate change scenarios, *Clim. Change* 35 (4) (1997) 397–414, <https://doi.org/10.1023/A:1005342632279>.
- [31] D. Silver, T. Hubert, J. Schrittwieser, I. Antonoglou, M. Lai, A. Guez, M. Lanctot, L. Sifre, D. Kumaran, T. Graepel, T. Lillicrap, K. Simonyan, D. Hassabis, A general reinforcement learning algorithm that masters chess, shogi, and go through self-play, *Science* 362 (6419) (2018) 1140–1144, <https://doi.org/10.1126/science.aar6404>.
- [32] P. Steduto, T.C. Hsiao, D. Raes, E. Fereres, AquaCrop—the FAO crop model to simulate yield response to water: i. concepts and underlying principles, *Agron. J.* 101 (3) (2009) 426–437, <https://doi.org/10.2134/agronj2008.0139s>.
- [33] R.S. Sutton, A.G. Barto, *Introduction to Reinforcement Learning*, 135, MIT press Cambridge, 1998. Vol.
- [34] USDA. (2019). National Agricultural Statistics Service—Charts and Maps—Prices Received: corn Prices Received by Month, US. https://www.nass.usda.gov/Charts_and_Maps/AgriculturalPrices/pricecn.php.
- [35] USDA-NASS. (2018). 2018 Irrigation and Water Management Survey. 3 (November).
- [36] O. Vinyals, I. Babuschkin, W.M. Czarnecki, M. Mathieu, A. Dudzik, J. Chung, D. H. Choi, R. Powell, T. Ewalds, P. Georgiev, J. Oh, D. Horgan, M. Kroiss, I. Danihelka, A. Huang, L. Sifre, T. Cai, J.P. Agapiou, M. Jaderberg, D. Silver, Grandmaster level in StarCraft II using multi-agent reinforcement learning, *Nature* 575 (7782) (2019) 350–354, <https://doi.org/10.1038/s41586-019-1724-z>.
- [37] D. Wang, X. Cai, Irrigation scheduling—role of weather forecasting and farmers' behavior, *J. Water Resour. Plann. Manage.* 135 (5) (2009) 364–372, [https://doi.org/10.1061/\(ASCE\)0733-9496\(2009\)135:5\(364\)](https://doi.org/10.1061/(ASCE)0733-9496(2009)135:5(364)).
- [38] Y. Yang, J. Hu, D. Porter, T. Marek, K. Heflin, H. Kong, Deep reinforcement learning-based irrigation scheduling, *Trans. ASABE* 63 (3) (2020) 549–556, <https://doi.org/10.13031/trans.13633>.
- [39] R. Young, T. Foster, T. Mieno, A. Valocchi, N. Brozović, Hydrologic-Economic trade-offs in groundwater allocation policy design, *Water Resour. Res.* 57 (1) (2021), <https://doi.org/10.1029/2020WR027941> e2020WR027941.
- [40] W. Zhao, J.P. Queralta, T. Westerlund, Sim-to-Real transfer in deep reinforcement learning for robotics: a survey, 2020 IEEE Symp. Series Comput. Intell. (2020) 737–744. SSCI 2020.